Strong cone avoidance for computably bounded progressive colorings of variable-sized finite sets

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Editorial note. The coloring f need not be Z-computable. The bounding function h is Z-computable; this distinction is essential to strong cone avoidance.

Editorial note. All new forcing conditions, stabilization/extension lemmas, relation coding, specification and diagonalization arguments are included in the solution. No forcing lemma is supplied as a free prerequisite.

Editorial note. The target is the progressive-coloring theorem itself. Its rainbow corollary and finite-dimensional special cases are not counted separately.

Editorial scope. One original result. Full source-local mathematical construction included; named classical inputs are explicit.

Allowed standard prerequisites (editorial).

- Finite-dimensional strong cone avoidance. Wang strong cone avoidance for fixed-arity free sets; Cholak-Patey strong cone avoidance for the achromatic Ramsey theorem at the Catalan bound C_n. Both cited classical statements are supplied verbatim.
- Effective compactness and countable forcing. König compactness for finitely branching trees, Sigma01(X) enumeration, and constructing a descending chain meeting countably many dense requirements.

There exist four other subsystems which, together with RCA_0, calibrate the strength of most theorems. These subsystems are known as the "Big Five" (see Montálban [2]). Among these, ACA_0 (Arithmetic Comprehension Axiom) extends the axioms of RCA_0 with the comprehension scheme for all arithmetic formulas. Based on the correspondence between computability and definability, it is equivalent to stating the existence of the Turing jump of any set ($\forall X \exists Y (Y = X')$).

We shall also consider two lesser-known stronger variants of ACA_0, namely, ACA_0' and ACA_0^+. The system ACA_0' extends RCA_0 with the axiom stating closure under all finite jumps; the system ACA_0^+ extends RCA_0 with the axiom stating the existence of the ω-jump of any set. The ω-jump of a set X is the set $X^{(\omega)} = \bigoplus_n X^{(n)}$. The system ACA_0^+ is famous for being the best known upper bound to Hindman's theorem.

Let us first fix some notation and recall the definition of these principles for colorings of finite subsets of fixed size. We denote by $\mathbb{N}$ the set of natural numbers and by $\mathbb{N}^+$ the set of positive natural numbers. For $X \subseteq \mathbb{N}$ and $n \in \mathbb{N}$

we denote by $[X]^n$ the set of all n-subsets of X. We denote by $[X]^{<\omega}$ the set of all finite subsets of X. We always assume that sets are presented in increasing order and make no distinction between a set and the sequence of its elements in increasing order. We write $x \cdot y$ for the concatenation of two sequences x and y. We identify a natural number with the set of its predecessors, so that, if $k \geq 1$ we can write $f : X \to k$ to declare a function from X to $\{0, 1, \dots, k-1\}$. A function of this type is often called a coloring of X in k colors.

Given a set X and $n \in \mathbb{N}$, we write $[X]^{\leq n}$ for the set of all subsets $s \subseteq X$ such that $|s| \leq n$, and $[X]^{\leq !\omega}$ for the set of all subsets $s \subseteq X$ such that $|s| \leq 1 + \min s$. By convention, for $n = 0$, $[X]^{\leq n}$ is the singleton $\{\emptyset\}$. Given a function $h : \mathbb{N} \to \mathbb{N}$, we write $[X]^{\leq h(\cdot)}$ for the set of all finite $s \subseteq X$ such that $|s| \leq h(\min s)$.

Definition 0.1. *A set $B \subseteq [\mathbb{N}]^{<\omega}$ is* ω-bounded *if and only if $B \subseteq [\mathbb{N}]^{\leq h(\cdot)}$ for some function $h : \mathbb{N} \to \mathbb{N}$. It is* computably ω-bounded *if furthermore h is computable.*

Definition 0.2. *A problem P admits* strong cone avoidance *if for every set Z, every non-Z-computable set C and every P-instance X, there exists a P-solution Y to X such that $C \not\leq_T Y \oplus Z$.*

Note that in the previous definition, no computability constraint is given on the P-instance X. Thus, strong cone avoidance reflects the combinatorial weakness of P, in the sense that no matter how complex the instance is, it cannot code in its solutions an infinite binary sequence. We shall use the following two theorems:

Theorem 0.3 (Wang [3])**.** *For every $n \in \mathbb{N}$, FS^n, TS^n and RRT^n admit strong cone avoidance.*

Definition 0.4 (Achromatic Ramsey Theorem)**.** *Let $n, k, \ell \in \mathbb{N}^+$. For every coloring $f : [\mathbb{N}]^n \to k$, there exists an infinite $H \subseteq \mathbb{N}$ such that $|f([H]^n)| \leq \ell$. We denote this statement $\mathsf{RT}^n_{k,\ell}$. We write $\mathsf{RT}^n_{<\infty,\ell}$ for $\forall k \mathsf{RT}^n_{k,\ell}$.*

The following sequence of numbers, known as *Catalan numbers*, is omnipresent in Combinatorics. It is inductively defined as follows:

$$C_0 = 1 \qquad C_{n+1} = \sum_{i=0}^{n} C_i C_{n-i}$$

This sequence starts with $1, 1, 2, 5, 14, 42, \dots$ (see sequence A000108 in the OEIS). The number C_n admits many characterizations, such as the number of ways of associating n applications of a binary operator. In computability theory, the nth Catalan number C_n surprisingly arose as the exact threshold ℓ at which $\mathsf{RT}^n_{<\infty,\ell}$ admits strong cone avoidance.

Theorem 0.5 (Cholak and Patey [1])**.** *For every $n \in \mathbb{N}^+$, $\mathsf{RT}^n_{<\infty,C_n}$ admits strong cone avoidance.*

Definition 0.6. *Let $B \subseteq [\mathbb{N}]^{<\omega}$ be a set and $f : B \to [\mathbb{N}]^{<\omega}$ be a coloring. A set $H \subseteq \mathrm{base}(B)$ is* f-free *if for every $s \in B \cap [H]^{<\omega}$, $f(s) \cap H \subseteq s$.*

For example, given a coloring $f : [\mathbb{N}]^n \to \mathbb{N}$, one can let $B = [\mathbb{N}]^n$ and $g : [\mathbb{N}]^n \to [\mathbb{N}]^{<\omega}$ be defined by $g(s) = \{f(s)\}$. Then a set is f-free if and only if it is g-free. Of course, even with $B = [\mathbb{N}]^n$, infinite free sets do not necessarily exist for arbitrary colorings. We need to impose some constraints on the size of the sets in the image of f.

Definition 0.7. *Fix $B \subseteq [\mathbb{N}]^{<\omega}$. A coloring $f : B \to [\mathbb{N}]^{<\omega}$ is* b-constrained *for a bounding function $b : \mathbb{N} \to \mathbb{N}$ if for every $s \in B$, $|f(s)| \leq b(\min s)$. If b is the constant function $x \mapsto k$, then we say that f is* k-constrained.

Definition 0.8 (k-Constrained Free Set Theorem)**.** *Fix $B \subseteq [\mathbb{N}]^{<\omega}$ and $k \in \mathbb{N}$. FS_k^B is the statement "For every coloring $f : B \to [\mathbb{N}]^{\leq k}$, there is an infinite f-free set".*

Given $n \in \mathbb{N}$ and a function $h : \mathbb{N} \to \mathbb{N}$, we write $\mathsf{FS}_k^{\leq n}$, $\mathsf{FS}_k^{\leq !\omega}$ and $\mathsf{FS}_k^{\leq h(\cdot)}$ for FS_k^B when B is $[\mathbb{N}]^{\leq n}$, $[\mathbb{N}]^{\leq !\omega}$ and $[\mathbb{N}]^{\leq h(\cdot)}$, respectively.

Definition 0.9. *Fix $B \subseteq [\mathbb{N}]^{<\omega}$. A function $f : B \to [\mathbb{N}]^{<\omega}$ is* progressive *if for every $s \in B$, either $f(s) = \emptyset$, or $\min f(s) \geq \min s$.*

An easy combinatorial argument shows that the following statement is classically true.

Definition 0.10 (Progressive Free Set Theorem)**.** *Fix a set $B \subseteq [\mathbb{N}]^{<\omega}$ and a bounding function $b : \mathbb{N} \to \mathbb{N}$. PFS_b^B is the statement "For every b-constrained progressive coloring $f : B \to [\mathbb{N}]^{<\omega}$, there is an infinite f-free set".*

Theorem 0.11. *Fix a set Z, a non-Z-computable set D and a Z-computable function $h : \mathbb{N} \to \mathbb{N}$. For every h-constrained progressive coloring $f : [\mathbb{N}]^{\leq h(\cdot)} \to [\mathbb{N}]^{<\omega}$, there exists an infinite f-free set $G \subseteq \mathbb{N}$ such that $D \not\leq_T G \oplus Z$.*

Proposition 0.12. *For every $k, n \in \mathbb{N}$, $\mathsf{FS}_k^{\leq n}$ admits strong cone avoidance.*

Proof. Fix some set Z, some non-Z-computable set C, and some coloring $f : [\mathbb{N}]^{\leq n} \to [\mathbb{N}]^{\leq k}$. For every $m \leq n$ and $j < k$, let $f_{m,j} : [\mathbb{N}]^m \to \mathbb{N}$ be the coloring defined for every $s \in [\mathbb{N}]^m$ by letting $f_{m,j}(s)$ be the jth element of $f(s)$, if it exists, and $f_{m,j}(s) = 0$ otherwise. By finitely many successive applications of strong cone avoidance of FS^m for $m \leq n$ (see Wang [3]), there is an infinite set $H \subseteq \mathbb{N}$ which is simultaneously $f_{m,j}$-free for every $m \leq n$ and $j < k$, and such that $C \not\leq_T H \oplus Z$.

We claim that H is f-free. Suppose by way of contradiction that there is some $s \in [H]^{\leq n}$ and some $c \in (f(s) \cap H) \setminus s$. Let j be such that c is the jth element of $f(s)$, and let $m = |s|$. Then $f_{m,j}(s) = c$, contradicting $f_{m,j}$-freeness of H. □

Proof. For simplicity, we prove the theorem in a non-relativized form. Relativization is straightforward. Fix a non-computable set D. Let $\mathcal{P}$ be the collection of all progressive colorings of type $[\mathbb{N}]^{\leq h(\cdot)} \to [\mathbb{N}]^{<\omega}$. For a function $b : \mathbb{N} \to \mathbb{N}$ we write $\mathcal{P}_b$ for the class of all b-constrained colorings in $\mathcal{P}$.

Given two colorings $f, g \in \mathcal{P}$, we write $g \leq f$ if for every $s \in [\mathbb{N}]^{\leq h(\cdot)}$, $g(s) \supseteq f(s)$. Note that if H is g-free and $g \leq f$, then H is f-free. Given two colorings $f, g \in \mathcal{P}$, let $f \cup g$ be the coloring defined by $(f \cup g)(s) = f(s) \cup g(s)$. The coloring $f \cup g$ is the greatest lower bound of f and g with respect to $\leq$.

Consider the following notion of forcing:

Definition 0.13. *A* condition *is a triple* (f, σ, X) *such that* $f \in \mathcal{P}$ *is* b_f*-constrained, for some computable function* $b_f : \mathbb{N} \to \mathbb{N}$, $\sigma \in [\mathbb{N}]^{<\omega}$, $X \subseteq \mathbb{N}$ *is an infinite set such that* $\max \sigma < \min X$, *and*

(a) for every $s \in [\sigma \cup X]^{\leq h(\cdot)}$ *with* $\min s \in \sigma$, $f(s) \cap X \subseteq s$.

(b) for every $s \in [\sigma \cup X]^{\leq h(\cdot)}$, $f(s) \cap \sigma \subseteq s$.

(c) $D \not\leq_T X$.

A condition $d = (g, \tau, Y)$ extends $c = (f, \sigma, X)$ *(written* $d \leq c$*) if* $g \leq f$, $\tau \succeq \sigma$, $Y \subseteq X$ *and* $\tau \setminus \sigma \subseteq X$.

The following lemma states that property (a) can be obtained "for free", that is, by restricting the reservoir, and therefore does not impose any constraint on the stem. In what follows, fix a coloring $f \in \mathcal{P}$ which is b_f-constrained, for some computable function $b_f : \mathbb{N} \to \mathbb{N}$.

Lemma 0.14. *For every* $\sigma \in [\mathbb{N}]^{<\omega}$ *and every infinite set* X *such that* $\max \sigma < \min X$ *and* $D \not\leq_T X$, *there is an infinite set* $Y \subseteq X$ *such that* $D \not\leq_T Y$ *and for every* $s \in [\sigma \cup Y]^{\leq h(\cdot)}$ *with* $\min s \in \sigma$, $f(s) \cap Y \subseteq s$.

Proof. For every $t \in [\sigma]^{\leq h(\cdot)}$ with $t \neq \emptyset$, let $f_t : [X]^{\leq h(\min t) - |t|} \to [\mathbb{N}]^{\leq b_f(\min t)}$ be defined for every u by $f_t(u) = f(t \cdot u)$. By finitely many successive applications of Proposition 0.12, there exists an infinite subset $Y \subseteq X$ such that $D \not\leq_T Y$ and such that Y is simultaneously f_t-free for every $t \in [\sigma]^{\leq h(\cdot)}$ with $t \neq \emptyset$.

We claim that Y is our desired set. Indeed, for every $s \in [\sigma \cup Y]^{\leq h(\cdot)}$ with $\min s \in \sigma$, letting $t = s \cap \sigma$ and $u = s \cap Y$, we have $f(s) = f_t(u)$, so by f_t-freeness of Y, $f(s) = f_t(u) \cap Y \subseteq u \subseteq s$. □

In what follows, recall that C_n stands for the nth Catalan number.

Definition 0.15. *A set* X stabilizes σ *if for every* $t \in [\sigma]^{\leq h(\cdot)}$ *with* $t \neq \emptyset$ *and every* $n \leq h(\min t) - |t|$, *there is a set* $I_{t,n} \subseteq [\sigma]^{\leq b_f(\min t) \times C_n}$ *such that for every* $u \in [X]^n$, $f(t, u) \cap \sigma \subseteq I_{t,n}$.

Lemma 0.16. *For every* $\sigma \in [\mathbb{N}]^{<\omega}$ *and every infinite set* X *such that* $\max \sigma < \min X$ *and* $D \not\leq_T X$, *there is an infinite set* $Y \subseteq X$ *stabilizing* σ *such that* $D \not\leq_T Y$.

Proof. For every $t \in [\sigma]^{\leq h(\cdot)}$ with $t \neq \emptyset$ and every $n \leq h(\min t) - |t|$, let $g_{t,n} : [X]^n \to [\sigma]^{\leq b_f(\min t)}$ be defined by $g_{t,n}(u) = f(t \cdot u) \cap \sigma$. By finitely many successive applications of strong cone avoidance of $\mathsf{RT}^n_{<\infty, C_n}$ (see Cholak and Patey [1]), there is an infinite subset $Y \subseteq X$ such that $D \not\leq_T Y$ and for every $t \in [\sigma]^{\leq h(\cdot)}$ with $t \neq \emptyset$ and every $n \leq h(\min t) - |t|$, $|g_{t,n}([Y]^n)| \leq C_n$. Note that here, a color is an element of $[\sigma]^{\leq b_f(\min t)}$ instead of a natural number. Let $I_{t,n} = \bigcup g_{t,n}([Y]^n)$. Then $|I_{t,n}| \leq b_f(\min t) \times C_n$. By definition, for every $u \in [Y]^n$, $f(t \cdot u) \cap \sigma = g_{t,n}(u) \subseteq I_{t,n}$. □

Given a computable function $b : \mathbb{N} \to \mathbb{N}$, let $b^+ : \mathbb{N} \to \mathbb{N}$ be the computable function defined by $b^+(m) = \sum_{n \leq h(m)} b(m) \times C_n$.

Definition 0.17. *Let* X *be a reservoir stabilizing* $[0, k]$. *The* limit coloring *is the function* $g_{k,X} : [k]^{\leq h(\cdot)} \to [k]^{<\omega}$ *defined by* $g_{k,X}(\emptyset) = \emptyset$ *and* $g_{k,X}(t) = \bigcup_{n \leq h(t) - |t|} I_{t,n}$ *otherwise.*

The limit function $g_{k,X}$ is b_f^+-constrained. Note that if $\rho \subseteq [0,k]$ is $g_{k,X}$-free, then it is f-free. Indeed, for every $t \in [0,k]^{\leq h(\cdot)}$, $g_{k,X}(t) = I_{t,0}$ and by definition of stability for $n = 0$, $f(t) \cap [0,k] \subseteq I_{t,0}$.

The following lemma is the core combinatorial lemma which specifies the conditions under which a block of elements ρ can be added to the stem while preserving the property (b).

Lemma 0.18. *Let (f, σ, X) be a condition and $Y \subseteq X$ be an infinite set stabilizing $[0,k]$ for some $k \in \mathbb{N}$ and let $g_{k,Y} : [k]^{\leq h(\cdot)} \to [k]^{<\omega}$ be the limit coloring. Let $\rho \subseteq X \restriction_k$ be a finite $g_{k,Y}$-free set. Then $(f, \sigma \cup \rho, Y)$ satisfies property (b).*

Proof. Fix some $s \in [\sigma \cup \rho \cup Y]^{\leq h(\cdot)}$. We have multiple cases.

- Case 1: $s \cap \sigma \neq \emptyset$. Then by properties (a) and (b) of (f, σ, X), $f(s) \cap (\sigma \cup X) \subseteq s$. Since $(\sigma \cup \rho) \subseteq (\sigma \cup X)$, then $f(s) \cap (\sigma \cup \rho) \subseteq s$.
- Case 2: $s \cap \sigma = \emptyset$ but $s \cap \rho \neq \emptyset$. By property (b) of (f, σ, X), $f(s) \cap \sigma \subseteq s$. Let $t = s \cap \rho$ and $u = s \cap Y$. Since ρ is $g_{k,Y}$-free, $g_{k,Y}(t) \cap \rho \subseteq t$. By definition of $g_{k,Y}(t)$, $I_{t,|u|} \subseteq g_{k,Y}(t)$, so $I_{t,|u|} \cap \rho \subseteq t$. In particular, $f(s) = f(t,u) \in I_{t,|u|}$, so $f(s) \cap \rho \subseteq t \subseteq s$. Thus, $f(s) \cap (\sigma \cup \rho) \subseteq s$.
- Case 3: $s \cap (\sigma \cup \rho) = \emptyset$. Then $s \subseteq Y$. Since $\min Y > \max(\sigma \cup \rho)$, then by progressiveness of f, $f(s) \cap (\sigma \cup \rho) = \emptyset$.

□

One can combine Lemmas 0.14, 0.16 and 0.18 to obtain an extensibility lemma, saying that every sufficiently generic filter induces an infinite set.

Lemma 0.19. *Let (f, σ, X) be a condition. There is an extension $(f, \tau, Y) \leq (f, \sigma, X)$ such that $|\tau| > |\sigma|$.*

Proof. Let $x = \min X$. By Lemma 0.16, there is an infinite subset $Y_0 \subseteq X$ stabilizing $[0,x]$ such that $D \not\leq_T Y_0$. Let g be the limit function. By Lemma 0.14, there is an infinite subset $Y \subseteq Y_0$ such that $(f, \sigma \cup \{x\}, Y)$ satisfies property (a). By Lemma 0.18, $\{x\}$ being vacuously g-free, $(f, \sigma \cup \{x\}, Y)$ satisfies property (b). Thus, $(f, \sigma \cup \{x\}, Y)$ is a valid extension. □

Given a computable function $b : \mathbb{N} \to \mathbb{N}$, the space $\mathcal{P}_b$ is not compact. However, $\mathcal{P}_b$ is in one-to-one correspondence with the effectively compact space $\mathcal{R}_b$ of all relations $R \subseteq [\mathbb{N}]^{\leq h(\cdot)} \times \mathbb{N}$ such that for every $s \in [\mathbb{N}]^{\leq h(\cdot)}$, $|\{y : (s,y) \in R\}| \leq b(\min s)$ and for every $(s,y) \in R$, $y \geq \min s$. Indeed, given a function $g \in \mathcal{P}_b$, one can define the relation $R_g \in \mathcal{R}_b$ defined by $R_g = \{(s,y) : y \in g(s)\}$ and given a relation $R \in \mathcal{R}_b$, the function $g_R \in \mathcal{P}_b$ is defined by $g_R(s) = \{y : (s,y) \in R\}$.

Note that the map $g \mapsto R_g$ is computable, but the map $R \mapsto g_R$ is not even continuous. Thankfully, there is a Turing functional which, given R and a finite set ρ, decides whether ρ is g_R-free or not. Indeed, to decide whether ρ is g_R-free, one does not need to know g_R restricted to $[\rho]^{\leq h(\cdot)}$, only to know R restricted to $[\rho]^{\leq h(\cdot)} \times \rho$.

We are now ready to define the forcing question:

Definition 0.20. *Let (f, σ, X) be a condition and $\varphi(G)$ be a Σ^0_1-formula. Let $(f, \sigma, X) ?\vdash \varphi(G)$ hold if and only if for every relation $R \in \mathcal{R}_{b^+_f}$, there is a finite g_R-free set $\rho \subseteq X$ such that $\varphi(\sigma \cup \rho)$ holds.*

The previous formulation of the forcing question is $\Pi^1_1(X)$ as it starts with a second-order universal quantification. However, thanks to the effective compactness of the space $\mathcal{R}_{b^+_f}$, it is equivalent to a $\Sigma^0_1(X)$-formula:

Lemma 0.21. *Let (f, σ, X) be a condition and $\varphi(G)$ be a Σ^0_1-formula. Then $(f, \sigma, X) ?\vdash \varphi(G)$ if and only if there is some $k \in \omega$ such that for every b^+_f-constrained progressive function $g : [0, k]^{\leq h(\cdot)} \to [0, k]^{<\omega}$, there is a finite g-free set $\rho \subseteq X \restriction_k$ such that $\varphi(\sigma \cup \rho)$ holds.*

Proof. Suppose first that there is some $k \in \mathbb{N}$ such that for every b^+_f-constrained progressive function $g : [0, k]^{\leq h(\cdot)} \to [0, k]^{<\omega}$, there is a finite g-free set $\rho \subseteq X \restriction_k$ such that $\varphi(\sigma \cup \rho)$ holds. Let $R \in \mathcal{R}_{b^+_f}$ be a relation, and let $g_R \in \mathcal{P}_{b^+_f}$ be the corresponding function. Then, letting $g : [0, k]^{\leq h(\cdot)} \to [0, k]^{<\omega}$ be defined by $g(s) = g_R(s) \cap [0, k]$, the function g is b^+_f-constrained and progressive, so there is some finite g-free set $\rho \subseteq X \restriction_k$ such that $\varphi(\sigma \cup \rho)$ holds. In particular, ρ is g_R-free. Since there is such a ρ for every $R \in \mathcal{R}_{b^+_f}$, then $(f, \sigma, X) ?\vdash \varphi(G)$ holds.

Suppose now that for every $k \in \mathbb{N}$, there is a b^+_f-constrained progressive function $g_k : [0, k]^{\leq h(\cdot)} \to [0, k]^{<\omega}$ such that for every g_k-free set $\rho \subseteq X \restriction_k$, $\varphi(\sigma \cup \rho)$ does not hold. Let $\mathcal{T}$ be the tree which, at level k, contains all such functions g_k, and which is ordered by the function extension relation. The tree $\mathcal{T}$ is finitely branching, so by König's lemma, there is an infinite path $g \in \mathcal{T}$. This path is a function $g \in \mathcal{P}_{b^+_f}$ such that for every finite g-free set $\rho \subseteq X$, $\varphi(\sigma \cup \rho)$ does not hold. Then the relation R_g witnesses that $(f, \sigma, X) ?\vdash \varphi(G)$ does not hold. □

Any filter $\mathcal{F}$ for this notion of forcing induces a set

$$G_{\mathcal{F}} = \bigcup \{\sigma : (g, \sigma, X) \in \mathcal{F}\}$$

We say that a condition p *forces* a formula $\varphi(G)$ if $\varphi(G_{\mathcal{F}})$ holds for every sufficiently generic filter $\mathcal{F}$ containing p. The following lemma states that the forcing question meets its specification.

Lemma 0.22. *Let $p = (f, \sigma, X)$ be a condition and $\varphi(G)$ be a Σ^0_1-formula.*

1. *If $p ?\vdash \varphi(G)$, then there is an extension of p forcing $\varphi(G)$.*
2. *If $p ?\nvdash \varphi(G)$, then there is an extension of p forcing $\neg\varphi(G)$.*

Proof. Suppose first $p ?\vdash \varphi(G)$ holds. By Lemma 0.21, there is some $k \in \mathbb{N}$ such that for every b^+_f-constrained progressive function $g : [0, k]^{\leq h(\cdot)} \to [0, k]^{<\omega}$, there is a finite g-free set $\rho \subseteq X$ such that $\varphi(\sigma \cup \rho)$ holds. By Lemma 0.16, there is an infinite subset $Y_0 \subseteq X$ stabilizing $[0, k]$ and such that $D \not\leq_T Y_0$. Let $g_{k,Y_0} : [0, k]^{\leq h(\cdot)} \to [0, k]^{<\omega}$ be the limit function. Note that g_{k,Y_0} is b^+_f-constrained and progressive, so there is a finite g_{k,Y_0}-free set $\rho \subseteq X$ such that $\varphi(\sigma \cup \rho)$ holds. By Lemma 0.18, $(f, \sigma \cup \rho, Y_0)$ satisfies (b). By Lemma 0.14, there

is an infinite subset $Y \subseteq Y_0$ such that $(f, \sigma \cup \rho, Y)$ satisfies (a) and $D \not\leq_T Y$. Thus $(f, \sigma \cup \rho, Y)$ is a valid extension of p. By choice of ρ, it forces $\varphi(G)$.

Suppose $p \,?\!\nvdash \varphi(G)$ holds. Then there is a relation $R \in \mathcal{R}_{b_f^+}$ such that for every finite g_R-free set $\rho \subseteq X$, $\varphi(\sigma \cup \rho)$ does not hold. Let $g_R \in \mathcal{P}_{b_f^+}$ be the corresponding function. Let $\hat{f} : [\mathbb{N}]^{\leq h(\cdot)} \to [\mathbb{N}]^{<\omega}$ be defined by $\hat{f}(s) = f(s) \cup (g_R(s) \setminus \sigma)$. Note that $\hat{f} \leq f$, and every $\hat{f}$-free subset $\rho \subseteq X$ is g_R-free. Moreover, $\hat{f}$ is $(b_f + b_f^+)$-constrained and progressive. Since (f, σ, X) satisfies (b) and for every $s \in [\sigma \cup X]^{\leq h(\cdot)}$, $\hat{f}(s) \cap \sigma \subseteq f(s) \cap \sigma \subseteq s$, then $(\hat{f}, \sigma, X)$ satisfies (b). By Lemma 0.14, there is an infinite subset $Y \subseteq X$ such that $D \not\leq_T Y$ and $(\hat{f}, \sigma, Y)$ satisfies (a). Thus $q = (\hat{f}, \sigma, Y)$ is a valid extension of p.

We claim that q forces $\neg\varphi(G)$. Indeed, suppose there is an extension $(\tilde{f}, \tau, Z) \leq q$ such that $\varphi(\tau)$ holds. Then by property (b) of $(\tilde{f}, \tau, Z)$, τ is $\tilde{f}$-free, and since $\tilde{f} \leq \hat{f}$, τ is $\hat{f}$-free. Let $\rho = \tau \setminus \sigma$. By definition of $\hat{f}$, ρ is a g_R-free subset of X, contradicting our choice of g_R. □

We can now prove our diagonalization lemma.

Lemma 0.23. *Let $p = (f, \sigma, X)$ be a condition and Φ_e be a Turing functional. There is an extension of p forcing $\Phi_e^G \neq D$.*

Proof. Let $U = \{(x, v) \in \mathbb{N} \times 2 : p \,?\!\vdash \Phi_e^G(x) \downarrow = v\}$. We have three cases:

- Case 1: $(x, 1 - D(x)) \in U$ for some $x \in \mathbb{N}$. By Lemma 0.22, there is an extension of p forcing $\Phi_e^G(x) \downarrow = 1 - D(x)$, hence forcing $\Phi_e^G \neq D$.
- Case 2: $(x, D(x)) \notin U$ for some $x \in \mathbb{N}$. By Lemma 0.22, there is an extension of p forcing $\neg(\Phi_e^G(x) \downarrow = D(x))$, hence forcing $\Phi_e^G \neq D$.
- Case 3: U is the graph of the characteristic function of D. By Lemma 0.21, the set U is $\Sigma_1^0(X)$, so $D \leq_T X$, contradiction.

□

We are now ready to prove Theorem 0.11. Let $f : [\mathbb{N}]^{\leq h(\cdot)} \to [\mathbb{N}]^{<\omega}$ be an h-constrained, progressive coloring, for a computable function $h : \mathbb{N} \to \mathbb{N}$. Let $\mathcal{F}$ be a sufficiently generic filter containing $(f, \emptyset, \mathbb{N})$, and let $G_{\mathcal{F}} = \bigcup\{\sigma : (g, \sigma, X) \in \mathcal{F}\}$. By definition of a forcing condition, $G_{\mathcal{F}}$ is f-free. By Lemma 0.19, $G_{\mathcal{F}}$ is infinite, and by Lemma 0.23, $D \not\leq_T G_{\mathcal{F}}$. This completes the proof of Theorem 0.11. □

References

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